# Redwan Ahmed Rizvee

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## Research Summary

Computer Science PhD student working at the intersection of **database systems and visual analytics**, with experience in machine learning and Python-based data analysis. Current research extends perception-aware sampling through density estimation, lighter preprocessing, and experimental evaluation of sampling quality and computational cost.

## Education

**University of Massachusetts Amherst** — Amherst, MA
*PhD in Computer Science* — Sep. 2025 – Present
Advisor: Dr. Alexandra Meliou | Anticipated graduation: May 2030
Research focus: perception-aware sampling, density estimation, and visual analytics.

**University of Dhaka** — Dhaka, Bangladesh
*MS in Computer Science and Engineering* — Jan. 2019 – Sep. 2021
CGPA: 4.00/4.00 | Ranked 1st in class

**University of Dhaka** — Dhaka, Bangladesh
*BSc in Computer Science and Engineering* — Jan. 2015 – Dec. 2018
CGPA: 3.85/4.00 | Ranked 1st in class | Dean's Award, Gold Medal

## Technical Skills

- **Data Analysis:** Python, SQL, NumPy, pandas, SciPy, scikit-learn, MATLAB; density estimation, clustering, sampling, and experimental evaluation.
- **Data Systems & Programming:** Vectorized batch processing, sampling synopses, PostgreSQL, MySQL; C++, C, Java, JavaScript, Shell.
- **Machine Learning:** PyTorch, OpenCV, reinforcement learning, graph-based optimization, transformer architectures, attention mechanisms, encoder-decoder models, saliency modeling.
- **Web & Tools:** Flask, Django, React, interactive visualization, Linux/Unix, Git, LaTeX.

## Research & Professional Experience

**University of Massachusetts Amherst (DREAM Lab)** — Amherst, MA
*Graduate Research Assistant* — Sep. 2025 – Present

- Extend PAwS with Python-based density estimation and lighter preprocessing, evaluating alternatives to saliency modeling to reduce computational cost while preserving sampling quality.
- Implement vectorized batch-processing workflows for sampling experiments; investigate runtime and memory use as part of an ongoing density-based formulation.
- Compare sampling methods across multiple datasets and sample sizes; enhance evaluation procedures to measure cluster separation and how well samples represent the original data.
- Independently built the **PAwS web demo**, enabling interactive comparisons of sampling methods and visual representations.

**TigerIT Bangladesh Limited** — Dhaka, Bangladesh
*Research Scientist* — July 2020 – Feb. 2022

- Researched audio noise reduction for plugins intended to improve live Google Meet calls, and worked on transliteration problems.
- Worked with transformer architectures and attention-based encoder-decoder models for audio and language-processing tasks.

**Data and Design Lab (DnD Lab)** — Dhaka, Bangladesh
*Research Assistant* — Apr. 2019 – Sep. 2019

- Collaborated with BRAC on the Bangladesh Quality of Life (**BDQOL**) indicator framework, developing data-processing pipelines and interactive dashboards for socio-economic data.

## Additional Research Experience

**Cognitive Agents and Interaction Lab (CAIL)** Dhaka, Bangladesh
*Co-Investigator* Jan. 2023 – Aug. 2025

- Supervised research applying deep learning and reinforcement learning to agent-based decision-making and human-in-the-loop systems.
- Developed **COLAS (Coding Lab Assistant)**, an LLM-assisted platform for source-code evaluation and student performance monitoring in an academic setting, incorporating code analysis and plagiarism detection.

**Data Mining Research Lab** Dhaka, Bangladesh
*Research Assistant* Jan. 2018 – Jul. 2025

- Developed tree-based sequential pattern mining algorithms, with an emphasis on candidate management, pruning, and computational efficiency.

## Selected Projects

**NavAble: Accessibility-Critical Perception** Spring 2026
*CS 682 Course Project, University of Massachusetts Amherst*

- Processed real and synthetic image datasets for accessibility-focused segmentation across 11 classes.
- Evaluated SegFormer, Mask2Former, and SAN across three training-data configurations; preliminary team results showed a 10.06-point mIoU gain for SegFormer with synthetic augmentation.

## Selected Publications & Manuscripts

- Z. Moumoulidou, H. Elhamdadi, **R. A. Rizvee**, K. Yang, S. Mitra, C. Xiong Bearfield, A. Meliou, "Perception-Augmented Databases for Effective Visual Analytics." **Manuscript under review for SIGMOD 2027; rebuttal submitted.**
- N. J. Lina, S. M. Mahmud, M. Hasan, M. F. Arefin, **R. A. Rizvee**, M. M. Rahman, M. M. Khan, "CodeStream: A dataset of iterative programming submissions with sequential verdict traces and attempt histories," *Data in Brief*, vol. 66, article 112843, 2026.
- **R. A. Rizvee**, C. F. Ahmed, C. K. Leung, "A tree-based framework to mine top-K closed sequential patterns," *Applied Intelligence*, vol. 55, article 221, 2025.
- **R. A. Rizvee**, C. F. Ahmed, M. F. Arefin, C. K. Leung, "A new tree-based approach to mine sequential patterns," *Expert Systems with Applications*, vol. 242, 122754, 2024.
- **R. A. Rizvee**, R. Hassan, M. M. Khan, "Reinforcement Learning-Based Formulations With Hamiltonian-Inspired Loss Functions for Combinatorial Optimization Over Graphs," *IEEE Access*, 2024.

## Posters & Research Mentorship

**New England Database Day (NEDB)** | *Poster Presenter* Jan. 2026
Presented perception-aware sampling research for large-scale visualization to the regional database research community. [Poster]

**LLMs as Visual Survey Agents for Scatter Plot Similarity** 2026
*PhD Mentor* | First author: Bach Nguyen
Mentored an undergraduate researcher in problem formulation and the research process for a project examining repeated similarity judgments, image-position effects, and opacity sensitivity using Gemini, Claude, and GPT. [Poster]

**Summer Undergraduate Research Program** Summer 2026
*Student Research Mentor* | Mentored one undergraduate student in the research process.

## Professional Service

**International Conference on Very Large Data Bases (VLDB 2026)** Boston, MA
*Student Volunteer* 2026

## Honors & Awards

- **Research Excellence Recognition**, University of Dhaka, 2025.
- **Dean's Award & Gold Medal**, Faculty of Engineering, University of Dhaka, 2019.
- **Fellowship Scholarship**, ICT Division, Bangladesh.